

OpenStax Calculus Volume 3

Section 4.3 Exercises: Partial Derivatives

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For the following exercises, calculate the partial derivative using the limit definitions only.

1. $\frac{\partial z}{\partial x}$ for $z = x^2 - 3xy + y^2$
2. $\frac{\partial z}{\partial y}$ for $z = x^2 - 3xy + y^2$

For the following exercises, calculate the sign of the partial derivative using the graph of the surface.

[Figure: A partial paraboloid with vertex at the origin and pointing up.]

3. $f_x(1, 1)$
4. $f_x(-1, 1)$
5. $f_y(1, 1)$
6. $f_x(0, 0)$

For the following exercises, calculate the partial derivatives.

7. $\frac{\partial z}{\partial x}$ for $z = \sin(3x) \cos(3y)$
8. $\frac{\partial z}{\partial y}$ for $z = \sin(3x) \cos(3y)$
9. $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for $z = x^8 e^{3y}$
10. $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for $z = \ln(x^6 + y^4)$
11. Find $f_y(x, y)$ for $f(x, y) = e^{xy} \cos(x) \sin(y)$.
12. Let $z = e^{xy}$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
13. Let $z = \ln(\frac{x}{y})$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
14. Let $z = \tan(2x - y)$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
15. Let $z = \sinh(2x + 3y)$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
16. Let $f(x, y) = \arctan(\frac{y}{x})$. Evaluate $f_x(2, -2)$ and $f_y(2, -2)$.
17. Let $f(x, y) = \frac{xy}{x-y}$. Find $f_x(2, -2)$ and $f_y(2, -2)$.

18. Find $\frac{\partial z}{\partial x}$ at $(0, 1)$ for $z = e^{-x} \cos(y)$. Evaluate the partial derivatives at point $P(0, 1)$.
19. Given $f(x, y, z) = x^3 y z^2$, find $\frac{\partial^2 f}{\partial x \partial y}$ and $f_z(1, 1, 1)$.
20. Given $f(x, y, z) = 2 \sin(x + y)$, find $f_x(0, \frac{\pi}{2}, -4)$, $f_y(0, \frac{\pi}{2}, -4)$, and $f_z(0, \frac{\pi}{2}, -4)$.
21. The area of a parallelogram with adjacent side lengths that are a and b , and in which the angle between these two sides is θ , is given by the function $A(a, b, \theta) = ba \sin(\theta)$. Find the rate of change of the area of the parallelogram with respect to the following:
 - (a) Side a
 - (b) Side b
 - (c) Angle θ
22. Express the volume of a right circular cylinder as a function of two variables:
 - (a) its radius r and its height h .
 - (b) Show that the rate of change of the volume of the cylinder with respect to its radius is the product of its circumference multiplied by its height.
 - (c) Show that the rate of change of the volume of the cylinder with respect to its height is equal to the area of the circular base.
23. Calculate $\frac{\partial w}{\partial z}$ for $w = z \sin(xy^2 + 2z)$.

Find the indicated higher-order partial derivatives.

24. f_{xy} for $z = \ln(x - y)$
25. f_{yx} for $z = \ln(x - y)$
26. Let $z = x^2 + 3xy + 2y^2$. Find $\frac{\partial^2 z}{\partial x^2}$ and $\frac{\partial^2 z}{\partial y^2}$.
27. Given $z = e^x \tan y$, find $\frac{\partial^2 z}{\partial x \partial y}$ and $\frac{\partial^2 z}{\partial y \partial x}$.
28. Given $f(x, y, z) = xyz$, find f_{xyy} , f_{yxy} , and f_{yyx} .
29. Given $f(x, y, z) = e^{-2x} \sin(z^2 y)$, show that $f_{xyy} = f_{yxy}$.
30. Show that $z = \frac{1}{2}(e^y - e^{-y}) \sin x$ is a solution of the differential equation $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$.
31. Find $f_{xx}(x, y)$ for $f(x, y) = \frac{4x^2}{y} + \frac{y^2}{2x}$.
32. Let $f(x, y, z) = x^2 y^3 z - 3xy^2 z^3 + 5x^2 z - y^3 z$. Find f_{xyz} .
33. Let $F(x, y, z) = x^3 y z^2 - 2x^2 y z + 3xz - 2y^3 z$. Find F_{xyz} .
34. Given $f(x, y) = x^2 + x - 3xy + y^3 - 5$, find all points at which $f_x = f_y = 0$ simultaneously.
35. Given $f(x, y) = 2x^2 + 2xy + y^2 + 2x - 3$, find all points at which $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$ simultaneously.
36. Given $f(x, y) = y^3 - 3yx^2 - 3y^2 - 3x^2 + 1$, find all points on f at which $f_x = f_y = 0$ simultaneously.
37. Given $f(x, y) = 15x^3 - 3xy + 15y^3$, find all points at which $f_x(x, y) = f_y(x, y) = 0$ simultaneously.

38. Show that $z = e^x \sin y$ satisfies the equation $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$.
39. Show that $f(x, y) = \ln(x^2 + y^2)$ solves Laplace's equation $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$.
40. Show that $z = e^{-t} \cos(\frac{x}{c})$ satisfies the heat equation $\frac{\partial z}{\partial t} = c^2 \left(\frac{\partial^2 z}{\partial x^2} \right)$.
41. Find $\lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x}$ for $f(x, y) = -7x - 2xy + 7y$.
42. Find $\lim_{\Delta y \rightarrow 0} \frac{f(x, y+\Delta y) - f(x, y)}{\Delta y}$ for $f(x, y) = -7x - 2xy + 7y$.
43. Find $\lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x}$ for $f(x, y) = x^2 y^2 + xy + y$.
44. Find $\lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x}$ for $f(x, y) = \sin(xy)$.
45. The function $P(T, V) = \frac{nRT}{V}$ gives the pressure at a point in a gas as a function of temperature T and volume V . The letters n and R are constants. Find $\frac{\partial P}{\partial V}$ and $\frac{\partial P}{\partial T}$, and explain what these quantities represent.
46. The equation for heat flow in the xy -plane is $\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$. Show that $f(x, y, t) = e^{-2t} \sin x \sin y$ is a solution.
47. The basic wave equation is $f_{tt} = f_{xx}$. Verify that $f(x, t) = \sin(x + t)$ and $f(x, t) = \sin(x - t)$ are solutions.
48. The law of cosines can be thought of as a function of three variables. Let x, y , and θ be two sides of any triangle where the angle θ is the included angle between the two sides. Then, $F(x, y, \theta) = x^2 + y^2 - 2xy \cos \theta$ gives the square of the third side of the triangle. Find $\frac{\partial F}{\partial \theta}$ and $\frac{\partial F}{\partial x}$ when $x = 2, y = 3$, and $\theta = \frac{\pi}{6}$.
49. Suppose the sides of a rectangle are changing with respect to time. The first side is changing at a rate of 2 in./sec whereas the second side is changing at the rate of 4 in./sec. How fast is the diagonal of the rectangle changing when the first side measures 16 in. and the second side measures 20 in.? (Round answer to three decimal places.)
50. A Cobb-Douglas production function is $f(x, y) = 200x^{0.7}y^{0.3}$, where x and y represent the amount of labor and capital available. Let $x = 500$ and $y = 1000$. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at these values, which represent the marginal productivity of labor and capital, respectively.
51. The apparent temperature index is a measure of how the temperature feels, and it is based on two variables: h , which is relative humidity, and t , which is the air temperature. $A = 0.885t - 22.4h + 1.20th - 0.544$. Find $\frac{\partial A}{\partial t}$ and $\frac{\partial A}{\partial h}$ when $t = 20^\circ\text{F}$ and $h = 0.90$.